

AHARNet: Adaptive Human Activity Recognition Model for Multimodal Consumer Electronics With Different Computation Resources

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Abstract—With the development of consumer electronics, enhancing smart terminal devices to better serve humans has received increasing attention. Human Activity Recognition (HAR) plays a crucial role in enhancing and improving human life on smart terminal devices, and has become a popular research topic. HAR models based on deep learning have demonstrated impressive performance due to their powerful feature extraction capabilities. As HAR has rich application scenarios, HAR models typically need to be deployed on devices with different computation resources. However, the existing HAR models lack the adaptability to different computing resources. To address this issue, this paper proposes an adaptive HAR model with resource-aware capability, AHARNet. Our model has multiple output layers and achieves adaptive inference through an early-exiting mechanism. AHARNet first generates early-exiting decisions based on the available computation resources of the device through a computation resource perceptron. Then during inference, AHARNet only performs partial layers to meet the device's resource constraints. To validate the effectiveness of our model, experiments are performed on three public datasets. The experimental results show that compared to the baseline architecture, AHARNet can dynamically reduce over 61% of computational overhead while demonstrating comparable accuracy.

Index Terms—Human activity recognition, adaptive architecture, early-exiting, multimodal data, consumer electronics.

I. INTRODUCTION

HUMAN activity recognition (HAR) can recognize activities based on human activity data and has become an important tool in health monitoring, motion tracking, and smart manufacturing [1], [2], [3]. With the development of consumer electronics, such as smart watches, smartphones, etc. [4], more and more intelligent consumer electronics

provide better deployment platforms and more application scenarios for HAR [5], [6]. At the same time, HAR applications also provide more valuable functions for consumer electronics [7], [8]. Therefore, HAR research for consumer electronics devices has become a trend.

Benefiting from the portability of wearable consumer electronics, human activity recognition applications can often be deployed on these wearable devices. The processing flow of the HAR task is shown in Fig. 1, which includes data collection, data preprocessing, feature engineering, and activity recognition. With the development of deep learning, HAR models based on deep learning techniques are widely applied in the feature engineering and recognition phases of HAR tasks to efficiently extract data features and recognize activities [9], [10], [11], [12]. HAR models based on deep learning can automatically extract features from the data, eliminating the reliance on experts required for manual feature selection in traditional methods. Due to the diverse applications of HAR, HAR models are commonly deployed on consumer electronics with different resource constraints. However, existing HAR models are typically computationally intensive and have fixed structures and sizes, which prevent them from adapting to varying computational resources. Therefore, it often necessitates designing and training corresponding models based on the resource constraints of different devices, leading to expensive model design and training costs.

Due to the lack of adaptability to different computation resources, the actual deployment and application of existing HAR models are hindered. To enhance the adaptability of models on devices with different resource constraints, an early-exiting mechanism has been proposed to adjust the model size [13], [14], [15]. In the early-exiting mechanism, the model executes only a subset of layers during inference to obtain output results, without executing all layers. This efficient computational approach has been widely applied in the field of computer vision. In the HAR task, only a small number of efforts have also adopted the early-exiting mechanism to enhance the adaptability of the models. For example, in [16], an adaptive CNN for energy-efficient HAR was proposed, where the early-exiting mechanism adjusts the number of CNN layers in the model to meet the energy constraints of different devices.

To improve the recognition accuracy, the Transformer model is more widely applied in the HAR because it

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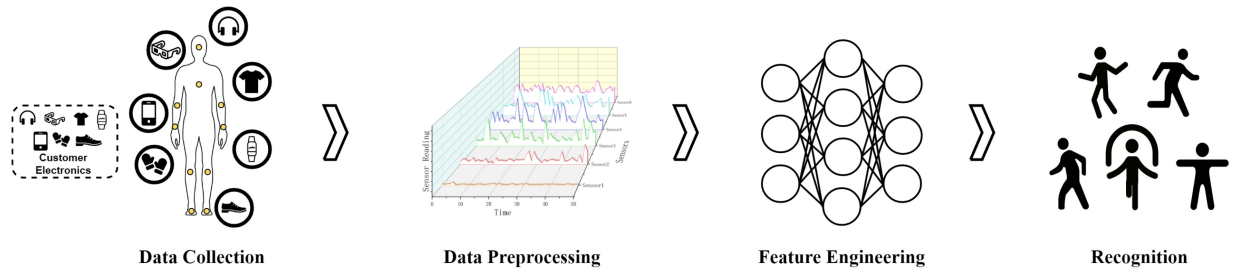


Fig. 1. Illustration of the process of human activity recognition based on wearable consumer electronics. There are four main stages in human activity recognition, which are data collection, data preprocessing, feature engineering and recognition.

demonstrates stronger feature extraction capabilities than LSTM. The Transformer, with its self-attention mechanism, can more fully capture the temporal correlations of HAR data and avoid performance degradation caused by long-distance dependencies. Therefore, in this paper, we adopt a CNN-Transformer hybrid model to extract spatial-temporal features from human activity data efficiently. Unlike the adaptive CNN layers proposed in [16], the primary overhead of the CNN-Transformer model lies in the Transformer layers. Therefore, to enhance the deployability of the HAR model based on CNN-Transformer on different devices, adaptive Transformer layers are more crucial.

Therefore, to improve the adaptability of the model to devices with different computation resources while ensuring high accuracy, this paper proposes an adaptive HAR neural network based on the early-exiting mechanism. In our experiments, AHARNet can reduce up to 61% of parameters, 65% of FLOPs, and 66% of runtime compared to the baseline architecture. The contributions of this paper are as follows.

- We propose AHARNet, an adaptive CNN-Transformer model with an early-exiting mechanism for human activity recognition, which improves adaptability while maintaining high recognition accuracy.
- We adopt a computation resource perceptron module to generate an early-exiting decision, which can be used to dynamically adjust the model size during inference. To the best of our knowledge, we are the first to consider computation resource awareness in adaptive HAR research.
- Experimental results on three public datasets show that AHARNet exhibits excellent recognition performance and adapts its size to reduce computational resource dependency.

The remainder of this paper is organized as follows. Section II reviews relevant work on the methods of HAR and adaptive deep learning techniques. Section III describes the proposed architecture of the AHARNet model and its details. Subsequently, the experimental setup and results are presented in Section IV. Finally, Section V concludes this paper.

II. RELATED WORK

A. Human Activity Recognition With Deep Learning

Human activity recognition is essentially a time series detection task. Due to the powerful feature extraction ability

of deep learning, the methods of HAR have shifted from traditional machine learning to deep learning [17]. The common deep learning modules in HAR tasks are CNN, LSTM, and Transformer. Reference [18] proposes an LSTM model based on bidirectional residual for HAR tasks, which effectively solves the problem of gradient disappearance in deep LSTM through residual connection. Reference [19] proposes a HAR model based on deep convolutional neural networks to extract features from accelerator and gyroscope data. Reference [20] proposes a HAR neural network consisting of convolutional layers and LSTM layers, which significantly improves the recognition accuracy. Due to the excellent performance of the Transformer model in sequence analysis, [21] and [22] propose the Transformer based HAR model.

To improve the recognition accuracy of the HAR task, the combination of CNN and the time series detection model has become a mainstream architecture. Compared with CNN-LSTM, the combination of CNN and Transformer can achieve better performance in HAR tasks [21], [22]. Therefore, to obtain better performance in HAR, the CNN-Transformer architecture is adopted in this paper. However, these advanced HAR models lack adaptivity to different computation resource constraints. So, it is necessary to combine other techniques to address this issue.

B. Adaptive Deep Learning

To improve the adaptability of deep models to different resource constraints, some research has employed the early-exiting mechanism for dynamic inference of models in various tasks [23]. References [24] and [25] proposed an early-exiting model based on classification confidence for image classification tasks. In these models, multiple output layers are placed after different depths of convolutional layers in computer vision models, and the decision that is used to terminate inference early is based on the prediction probability of the output layers. If the classification confidence of the shallow output layers exceeds a threshold, the classification result is outputted, and the model exits; otherwise, inference continues. In natural language processing (NLP) tasks, DeeBERT [26] uses the early dropout mechanism to speed up inference in the large-scale natural language model. ScaleFL [27] is a resource-adaptive federated learning method in heterogeneous client scenarios. It first utilizes the early-exiting mechanism to dynamically adjust the width and depth of the model to find the most suitable model for different client resource

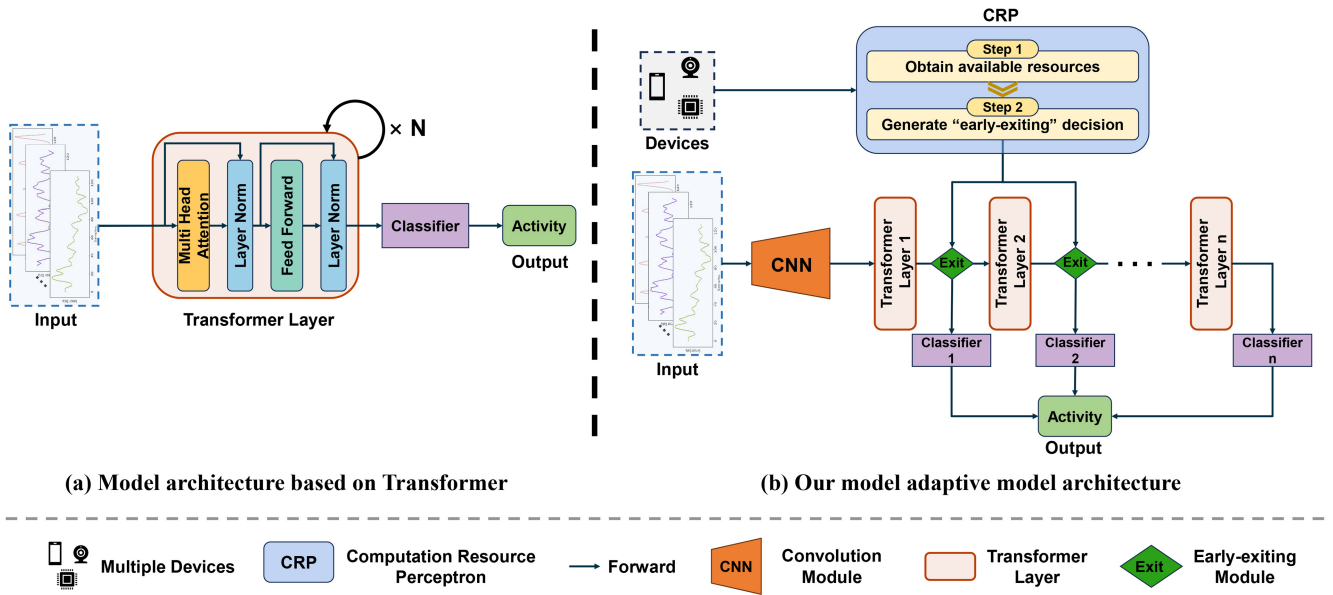


Fig. 2. Model architecture. (a) The baseline model architecture based on Transformer. (b) The architecture of the adaptive human activity recognition network.

constraints and then aggregates weights across multiple clients. Reference [16] proposed an energy-efficient adaptive CNN model based on early-exiting for HAR tasks, using an output block predictor to decide whether to perform early detection.

Although the previous early-exiting strategy based on classification confidence can improve the adaptability of the model to a certain extent, these methods cannot dynamically adjust the model according to the available computation resources of the device [25], [28]. In particular, when shallow output layers get low-confidence results, the model has to continue to perform subsequent computations, which is not friendly for small devices with limited resources. In addition, since HAR models usually need to be deployed on different types of devices, it is very valuable to improve the adaptability of HAR models. Therefore, to improve the resource-aware ability of the HAR model, this paper proposes an early-exiting mechanism based on resource-awareness and applies it to the high-performance HAR model.

III. METHODOLOGY

In order to improve the adaptability of the model to devices with different computation resources on HAR tasks, this paper proposes a resource-aware adaptive HAR model (AHARNet) based on the early-exiting mechanism. In this section, we present AHARNet in detail.

A. Motivation

In HAR application scenarios, HAR models are usually deployed to resource-limited devices, which makes the HAR model not only need to maintain high accuracy but also need to meet device resource constraints. Many existing deep learning models [29], [30] have demonstrated that improvements in accuracy usually benefit from an increase in model size. However, on resource-constrained devices, the model's depth or width cannot be increased indefinitely to enhance accuracy,

as this would exceed the available resources. Therefore, the motivation for designing models on resource-constrained devices is to maximize model size while satisfying device resource constraints to improve the accuracy of the model, as shown in equation (1).

$$\begin{aligned} \max \quad & S(\mathcal{M}) \\ \text{s.t.} \quad & R(\mathcal{M}) \leq C \end{aligned} \quad (1)$$

where $\mathcal{M}$ denotes the neural network model, $S(\cdot)$ denotes the model size, $R(\cdot)$ represents the computational resources required by the model, and C denotes the computation resource constraint of the device.

However, in a HAR application scenario consisting of a variety of different devices, there are often differences in the computation resources that these devices have. Since each device operates independently, different models of varying sizes must be designed to meet the specific resource constraints of each device. Customizing models for different devices is expensive and laborious. To solve this problem, we propose an adaptive HAR model architecture. The model allows each device to independently adjust its size during inference based on available computational resources, ensuring it meets resource constraints.

B. Model Architecture

The overall architecture of the proposed adaptive HAR network is illustrated in Fig. 2. AHARNet is comprised of four components: a computation resource perceptron (CRP), a CNN module, Transformer layers, and classifiers. Compared to traditional Transformer architectures with fixed sizes, our model incorporates multiple classifiers at different layer depths, enabling dynamic adjustment of the model size based on the device's available resources. Suppose our model consists of one CNN module, n Transformer layers, and n classifiers.

Based on the early-exiting mechanism, our model can dynamically select different numbers of feature extraction layers to form n submodels of varying sizes ($\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_n$), where submodel $\mathcal{M}_i$ consists of one CNN module, i Transformer layers (from Transformer layer 1 to i), and the i -th classifier. During the training process, we use a sequential approach to train submodels from small to large. This approach not only improves training speed but also optimizes submodel performance. During testing, AHARNet uses the computation resource perceptron (CRP) to access the device's hardware to obtain available computational resources and generate an early-exiting decision. Then, AHARNet exits feature extraction for the corresponding transformer layer according to the early-exiting decision. In other words, AHARNet selects the largest-size submodel while satisfying the resource constraint. Finally, the recognition results are output by the classifier in this submodel. Next, we will describe the convolution module, Transformer layers, classifiers, and computation resource perceptron in detail.

Convolution Module: Due to the effective spatial feature extraction capability, convolution operations can be employed to capture correlations among HAR data from different sensors. In our model, the convolution module consists of multiple convolution blocks and a downsampling layer. Each convolution block contains a 1D convolution layer (*Conv1d*), a batch normalization layer (*BN*), and a ReLU activation layer (*ReLU*). Equation (2) illustrates the forward propagation process within a convolution block, where X denotes the input HAR data, and Z_{conv} denotes the output of a convolution block.

$$Z_{conv} = \text{ReLU}(\text{BN}(\text{Conv1d}(X))) \quad (2)$$

After the execution of the convolution blocks, downsampling is applied to the output Z_{conv} to reduce the dimensionality by half as shown in equation (3), mitigating the computational load associated with high-dimensional features.

$$Z_{pool} = \text{MaxPooling}(Z_{conv}) \quad (3)$$

where Z_{pool} denotes the output of the MaxPooling layer.

Transformer Layer: Transformer excels in capturing long-term dependencies in time-series data and extracting interactions among different data. Therefore, our model adopts the Transformer layer to extract temporal features from HAR data. Our model comprises multiple standard Transformer layers. Each Transformer layer consists of a multi-head attention layer (*MHA*), a feedforward layer (*FF*), and two layer-normalization layers (*LN*). The forward propagation process of the Transformer layer is depicted in equation (4), where Z_0 denotes the input features of the Transformer layer, Z_1 denotes the intermediate features of the Transformer layer, Z_{trans} denotes the output of the Transformer layer.

$$\begin{aligned} Z_1 &= \text{LN}(Z_0 + \text{MHA}(Z_0)) \\ Z_{trans} &= \text{LN}(Z_1 + \text{FF}(Z_1)) \end{aligned} \quad (4)$$

Classifier: To ensure that the model possesses both adaptivity and high accuracy simultaneously, we have incorporated multiple classifiers within the model. Each classifier is individually connected to different depths of the network, allowing

control over the computational load of the model. Once the CRP module obtains information about the device's computational capabilities, the model adaptively selects the most suitable classifier for inference based on the optimization objective in equation (1). The classifier comprises a fully connected layer (*FC*), and the forward propagation process is depicted in equation (5), where $\hat{Y}$ denotes the prediction result. Each classifier has independent parameters, maximizing the accuracy of different submodels to the greatest extent possible.

$$\hat{Y} = \text{FC}(Z_{trans}) \quad (5)$$

Computation Resource Perceptron (CRP): The task of CRP is to dynamically adjust the model size at inference according to the available computational resources of the device. CRP first obtains the computational resources available on the device, and then generates an early-exiting decision to select an appropriate number of layers to form a submodel that meets the device's computational resource constraints for inference. Once the design of each layer in the model is finalized, we can obtain the computational resources needed for each layer. Consequently, the CRP ensures that the submodels generated based on the early-exiting decision comply with the device's resource constraints.

C. Model Training and Inference

The training and inference processes are outlined in Algorithm 1. Considering that on resource-constrained end devices, the lightest submodel $\mathcal{M}_1$ tends to have more opportunities to be utilized. Therefore, during training, we use a serial approach to train sub-models from small to large. This helps ensure that model weights are better adapted to the smaller sub-models, optimizing their performance. The training schematic is illustrated in Fig. 3. Initially, the shallowest model $\mathcal{M}_1$ is trained, consisting of the CNN block and a Transformer layer1, and classifier1. After the completion of training $\mathcal{M}_1$, the training of $\mathcal{M}_2$ begins. Since the $\mathcal{M}_2$ model includes $\mathcal{M}_1$, during the training of $\mathcal{M}_2$, the weights of $\mathcal{M}_1$ will be fixed, and only the newly added Transformer layer2 and classifier2 will be trained. Similarly, when training $\mathcal{M}_n$, the weights of $\mathcal{M}_{n-1}$ are fixed. When training the shallow submodels, the modules belonging to the deep submodel need not be involved. Through iterative training, we ensure that different submodels can share some weights, thereby reducing the overall parameters. As the HAR task is fundamentally a classification task, the cross-entropy loss is employed during the training phase, as shown in equation (6).

$$L_{CE}(Y, \hat{Y}) = - \sum_i^K Y_i \log(\hat{Y}_i) \quad (6)$$

where L_{CE} denotes the cross-entropy loss function, K denotes the number of classes in the task, Y denotes the true class label, and $\hat{Y}$ represents the predicted class by the model.

Once the model is trained, the entire AHARNet will be deployed on the device. During inference, the CRP module senses the available computational resources and generates early-exiting decisions to dynamically select a sub-model that meets the resource constraints for inference. When multiple

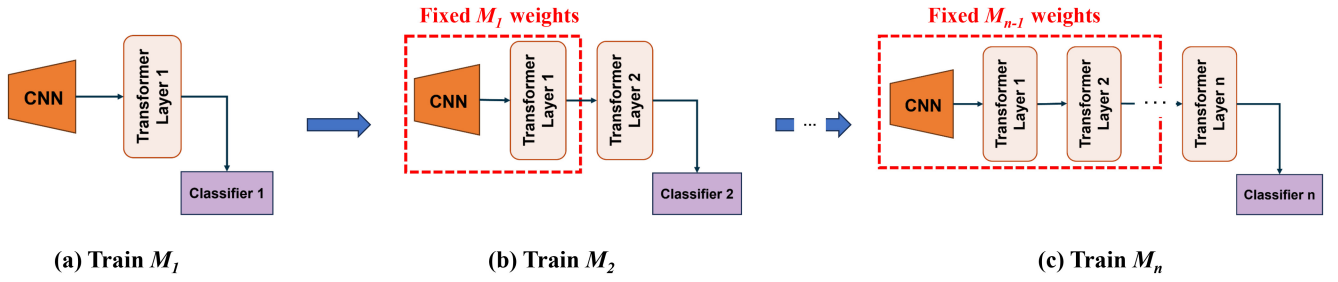


Fig. 3. Model training process. We train different layers sequentially from shallow to deep depending on the position of the classifier. When the shallow layers are trained, their weights will be fixed.

Algorithm 1 The Training and Testing Process of AHARNet

Require: X : Sensor dataset, Y : Activity category labels.

Ensure: $\hat{Y}$: Prediction results.

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1: Preprocess the dataset  $\{X, Y\}$ .
2:  $X_{train}, Y_{train}, X_{test}, Y_{test} \leftarrow$  Split the dataset  $X, Y$  into
   training and testing datasets.
3: if Training then
4:   Set the hyperparameters of the model.
5:   for  $M_i$  in  $\{M_1, M_2, \dots, M_n\}$  do
6:     if  $i > 1$  then
7:       Fix the weights of the parameters in  $M_{i-1}$ .
8:     end if
9:     Set the parameters of  $M_i$  to be trainable.
10:    for epoch in Epochs do
11:       $\hat{Y} = M_i(X_{train})$ .
12:      Compute the loss  $L_{CE}(Y_{train}, \hat{Y})$ .
13:      Backpropagate and update the parameters.
14:    end for
15:  end for
16: end if
17: if Testing then
18:   The CRP module selects the most suitable model  $M_j$ 
   based on the device's computation resource.
19:   Use model  $M_j$  for inference,  $\hat{Y} = M_j(X_{test})$ .
20:   Output the model's prediction results  $\hat{Y}$ .
21: end if

```

sub-models meet the device's computational resource constraints, the CRP module selects the deepest sub-model that satisfies the computational requirements, rather than running multiple models in parallel. This is because the largest sub-model typically provides stronger performance compared to other sub-models that meet the resource constraints [29], [30], making it unnecessary to waste resources on executing other sub-models. The selected sub-model will perform activity recognition based on the data collected by the sensors. Since our model consolidates multiple submodels into a single master model, there is no need to separately store multiple independent submodels.

IV. EXPERIMENTS

To validate the effectiveness of the proposed AHARNet, we conducted experiments on three popular HAR datasets: PAMAP2 [31], OPPORTUNITY [32], and Skoda [33]. We

TABLE I
CONFIGURATION FOR DIFFERENT SUBMODEL WITHIN AHARNet

Module	M_1	M_2	M_3
Convolution	$\begin{Bmatrix} Conv1d \\ BN \\ ReLU \end{Bmatrix} * 2$	$\begin{Bmatrix} Conv1d \\ BN \\ ReLU \end{Bmatrix} * 2$	$\begin{Bmatrix} Conv1d \\ BN \\ ReLU \end{Bmatrix} * 2$
MaxPooling(2)			
Transformer	$\begin{Bmatrix} MHA \\ LN \\ FF \\ LN \end{Bmatrix} * 1$	$\begin{Bmatrix} MHA \\ LN \\ FF \\ LN \end{Bmatrix} * 2$	$\begin{Bmatrix} MHA \\ LN \\ FF \\ LN \end{Bmatrix} * 3$
Classifier	$\{FC\}$	$\{FC\}$	$\{FC\}$

compared our method with various algorithms. Since the primary goal of our approach is to enhance the adaptivity of the model, we predefined three different classifiers. Therefore, our model can be split into three submodels with different sizes: M_1 , M_2 , and M_3 . Our model is implemented in PyTorch and executed on a CPU i7-11700 and GPU RTX 3060. We trained our models using the cosine optimizer.

A. Model Configuration

In this experiment, the different structures of submodels within AHARNet are shown in Table I. All submodels share the same convolution block. As the depth of the submodel increases, the number of layers in the Transformer layer increases. To maximize the reduction in storage cost, Transformer layers can be reused. Each submodel has its classifier to maximize model performance. During model training, the hyperparameters are set as follows: learning rate of 0.01, momentum of 0.9, weight decay of 0.0005, and dropout of 0.5.

B. Datasets

Three datasets are adopted to comprehensively verify the performance of the proposed model. These datasets consist of data collected by a variety of sensors. In this experiment, the number of sensors, sensor type, and data dimensions involved in each dataset are shown in Table II.

PAMAP2: In PAMAP2, there are 18 classes. The sampling frequency is 100Hz. For data preprocessing, we applied a standard dataset preprocessing method. We extracted data for 12 classes and used a sliding window size of 171 with a data overlap of 50%. During training on the PAMAP2 dataset, the batch size was set to 32, and each model was trained for 50 epochs.

TABLE II
THE NUMBER OF SENSORS, SENSOR TYPE, AND DATA DIMENSIONS IN EACH DATASET, WHERE A DENOTES ACCELEROMETER, G DENOTES GYROSCOPE, AND M DENOTES MAGNETOMETER

	PAMAP2	OPPORTUNITY	Skoda
Number of Sensors	4	56	20
Sensor Type	A,G	A,G,M	A
Data Dimension	36	113	60

OPPORTUNITY: In OPPORTUNITY, the sampling frequency is 30Hz. For data preprocessing, we focused on the 17 mid-level gesture actions and used a sliding window size of 30 with a data overlap of 50%. During training on the OPPORTUNITY dataset, the batch size was set to 128, and each model was trained for 20 epochs.

Skoda: In Skoda, there are 10 activities and the sampling frequency is 98Hz. For data preprocessing, we utilized the data from the right arm and applied a sliding window size of 98 with a data overlap of 10%. During training on the Skoda dataset, the batch size was set to 64, and each model was trained for 50 epochs.

C. Evaluation Metrics

To evaluate the effectiveness of the proposed model in this paper, we adopt four evaluation metrics, namely Macro F1-score, Parameters, FLOPs, and Runtime. Macro F1-score is used to evaluate the performance of the model, and the remaining three metrics are used to evaluate the adaptive ability of the proposed model.

Macro F1-score: In human activity recognition tasks, there is typically a significant imbalance in the number of samples across different activity classes. Therefore, the Macro F1-score is commonly used as a performance evaluation metric to better showcase the model's performance [22], [34]. This paper adopts the same Macro F1-score calculation method as existing algorithms,

$$\begin{aligned}
 F1 - Score &= \frac{1}{N} \sum_i^N 2 * \frac{Pre_i * Rec_i}{Pre_i + Rec_i} \\
 Pre_i &= \frac{TP_i}{TP_i + FP_i} \\
 Rec_i &= \frac{TP_i}{TP_i + FN_i} \\
 Acc_i &= \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i} \quad (7)
 \end{aligned}$$

where N denotes the total number of classes, Pre_i , Rec_i , and Acc_i denote the precision, recall, and accuracy for each class, respectively, TP , TN , FP , and FN denote true positive, true negative, false positive and true negative. In addition to the Macro F1-score, we also evaluate the precision, recall, and accuracy of different submodels of AHARNet in our experiments.

Parameters: This metric represents the number of model parameters and provides a measure of how much memory is required when the model is loaded.

FLOPs: This metric is used to count and measure the number of floating-point operations spent on model inference.

Runtime: The runtime of a model is the time it takes for the model to complete inference and is used to measure the computational efficiency of the model.

D. Experimental Results

Model Performance Results: The proposed AHARNet in this paper adopts a CNN-Transformer architecture. To comprehensively compare the performance of the AHARNet, we compared it with various structural models, such as LSTM [35], [36], CNN [19], [34], ConvLSTM [20], Attention [37], [38], [39], and Transformer [22]. The experimental results are shown in Table III, where $\Delta F1$ represents the difference in Macro F1-Score. $\Delta F1$ is calculated as the accuracy of AHARNet minus the accuracy of the comparison model.

Overall, the proposed adaptive HAR neural network (AHARNet) in this paper demonstrates high accuracy across all HAR datasets. As the model size increases, incorporating more Transformer layers, the performance of the model improves. On the PAMAP2 and Skoda datasets, AHARNet exhibits similarly outstanding performance compared to other models. On these two datasets, when compared to the traditional LSTM model, the accuracy of the most lightweight model of AHARNet- $\mathcal{M}_1$, increases by 18% and 2%, respectively. Compared to ConvLSTM+Attn incorporating attention mechanisms, the accuracy of our $\mathcal{M}_1$ model increases by 5% and 12%. Furthermore, on the OPPORTUNITY datasets, our model outperforms other models significantly. Compared to CNN and Transformer Encoder, the accuracy of $\mathcal{M}_1$ increases by 16% and 14%, respectively. Although the performance of $\mathcal{M}_1$ may be slightly lower than some methods, $\mathcal{M}_3$ can effectively improve the model accuracy. Compared with other methods, it can be seen that our model shows comparable performance with the existing methods under the premise of ensuring adaptation. This indicates that our model can better balance the computational overhead and model performance.

To provide a comprehensive evaluation of the proposed model's performance, the different size submodels were evaluated on various metrics, including F1-Score, accuracy, precision, and recall. As shown in Fig. 4, both F1-Score and accuracy increase with the deepening of the model layers. This indicates that deeper models possess stronger representation capabilities and are more adept at capturing distinguishing features among the data.

Model Adaptation Results: Table IV presents the number of parameters, FLOPs, and runtime for different model sizes on the three datasets. These three metrics are closely related to the structure of the model and the dimensions of the data fed to the model. In our experiments, parameters and FLOPs are the result of the model completing one input data. The runtime refers to the time required for the model to complete one batch during inference.

Since traditional HAR models typically have fixed sizes, according to [16], the $\mathcal{M}_3$ submodel can be regarded as the fixed-size baseline in our experiments. From Table IV, on the three datasets, it can be observed that compared to the $\mathcal{M}_3$, our $\mathcal{M}_1$ and $\mathcal{M}_2$ submodels have average parameter

TABLE III
MACRO F1-SCORE OF THE PROPOSED MODEL AND OTHER MODELS

Method	PAMAP2				OPPORTUNITY				Skoda			
	F1	$\Delta F1$			F1	$\Delta F1$			F1	$\Delta F1$		
		$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$		$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$		$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$
LSTM	0.75	0.18	0.19	0.20	0.63	0.12	0.13	0.13	0.89	0.02	0.04	0.05
Bi-directional LSTMs	0.84	0.09	0.1	0.11	0.68	0.07	0.08	0.08	0.91	0	0.02	0.03
Deep CNN	0.82	0.11	0.12	0.13	0.59	0.16	0.17	0.17	0.85	0.06	0.08	0.09
ConvAE	0.8	0.13	0.14	0.15	0.72	0.03	0.04	0.04	0.79	0.12	0.14	0.15
ConvLSTM	0.75	0.18	0.19	0.2	0.67	0.08	0.09	0.09	0.91	0	0.02	0.03
ConvLSTM+Attn	0.88	0.05	0.06	0.07	0.71	0.04	0.05	0.05	0.79	0.12	0.14	0.15
LSTM+ConAttn	0.9	0.03	0.04	0.05	-	-	-	-	0.94	-0.03	-0.01	0
AttnSense	0.89	0.04	0.05	0.06	0.66	0.09	0.1	0.1	0.93	-0.02	0	0.01
Transformer Encoder	0.95	-0.02	-0.01	0	0.61	0.14	0.15	0.15	0.93	-0.02	0	0.01
AHARNet-$\mathcal{M}_1$	0.93	-	-	-	0.75	-	-	-	0.91	-	-	-
AHARNet-$\mathcal{M}_2$	0.94	-	-	-	0.76	-	-	-	0.93	-	-	-
AHARNet-$\mathcal{M}_3$	0.95	-	-	-	0.76	-	-	-	0.94	-	-	-

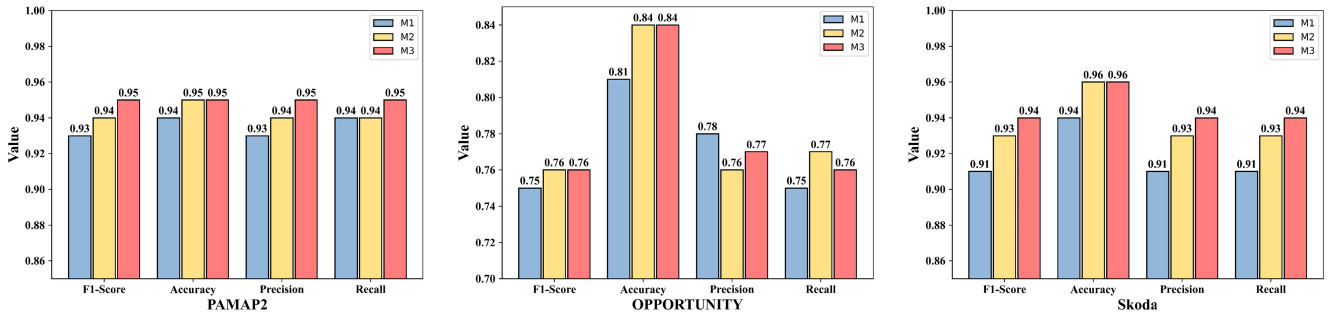


Fig. 4. F1-score, accuracy, precision, and recall for AHARNet with different sizes.

TABLE IV
PARAMETERS, FLOPS, AND RUNTIME FOR AHARNET
WITH DIFFERENT SIZES

	Method	PAMAP2	OPPORTUNITY	Skoda	Average Reduction
Parameters	$\mathcal{M}_1$	0.422M	0.092M	0.242M	61%
	$\mathcal{M}_2$	0.773M	0.155M	0.445M	31%
	$\mathcal{M}_3$ (baseline)	1.123M	0.219M	0.648M	-
FLOPs	$\mathcal{M}_1$	23.5M	4.3M	13.7M	65%
	$\mathcal{M}_2$	45.8M	8.2M	26.5M	33%
	$\mathcal{M}_3$ (baseline)	68.1M	12.2M	39.4M	-
Runtime	$\mathcal{M}_1$	0.273s	0.072s	0.174s	66%
	$\mathcal{M}_2$	0.528s	0.140s	0.335s	35%
	$\mathcal{M}_3$ (baseline)	0.844s	0.210s	0.505s	-

reductions of 61% and 31%, average FLOPs reductions of 65% and 33%, and average runtime reductions of 66% and 35%, respectively. This indicates that different submodels have varying computation resource requirements, allowing for dynamic adjustment of model size to meet the resource constraints of different devices. Furthermore, we report the parameter counts of the CNN and Transformer layers in our model, as shown in Table V. The Transformer layer has more parameters than the CNN layer, which demonstrates that our model can reduce its size more effectively by dynamically adjusting the number of Transformer layers.

Model Creation Overhead: Our model can dynamically load models of varying sizes based on the device's available resources, which necessitates consideration of the time required for model creation. Table VI presents the creation

TABLE V
THE NUMBER OF PARAMETERS AT DIFFERENT LAYERS OF THE MODEL
ACROSS DIFFERENT DATASETS

Method	PAMAP2	OPPORTUNITY	Skoda
Conv Layer	0.036M	0.015M	0.020M
Transformer Layer	0.351M	0.063M	0.203M

TABLE VI
MODEL CREATION TIME (IN MILLISECONDS) FOR MODELS OF VARYING
SIZES ACROSS DIFFERENT DATASETS

Method	PAMAP2	OPPORTUNITY	Skoda
$\mathcal{M}_1$	5.0ms	3.2ms	3.5ms
$\mathcal{M}_2$	9.4ms	4.5ms	6.1ms
$\mathcal{M}_3$	14.1ms	5.4ms	10.5ms

times of sub-models of varying sizes across different datasets. As shown in Table VI, the creation time increases progressively with model size. However, the creation time is in the millisecond range, which is significantly shorter than the model's runtime shown in Table IV. Therefore, the model creation time is negligible.

E. Discussion

The AHARNet proposed in this paper can be deployed on inference devices with different computation resources, enhancing the adaptability of HAR applications. Although the accuracy of smaller-sized submodels in the model slightly

decreases, for resource-constrained devices, achieving flexible deployment and efficient inference while meeting certain accuracy requirements is more crucial. Therefore, we prioritize the availability of the model within acceptable accuracy.

It is worth noting that the resource-aware adaptive model architecture proposed in this paper is a general framework. Leveraging the scalability of deep learning models, the proposed model can flexibly adjust the number of layers based on the accuracy requirements of different application scenarios. Furthermore, the model can be combined with existing advanced model optimization techniques to meet the specific requirements of particular application scenarios. For example, to further enhance the deployability of the model on resource-constrained devices, our model can incorporate excellent lightweight techniques.

From an energy efficiency perspective, the smaller submodels in our architecture require fewer computational resources than larger submodels, which theoretically enhances energy efficiency. However, our paper primarily focuses on enhancing model usability and does not specifically optimize for energy efficiency. Since energy efficiency has become the focus of various research fields at present [16], [40], [41], how to further improve the energy efficiency of our model needs further research.

V. CONCLUSION

In this paper, we propose an adaptive neural network for human activity recognition, AHARNet. The model uses the CNN-Transformer architecture to improve the feature extension ability and adopts the early-exiting strategy based on resource awareness to adjust the model size dynamically. AHARNet has multiple output layers that can be combined into multiple submodels of different sizes. AHARNet first obtains the available computation resource in the device through the CRP module. Then, the submodel with the highest accuracy that satisfies the resource constraint is selected for inference. In our experiments, we designed three submodels with different sizes, where the largest size served as the baseline architecture. The experimental results show that compared with the baseline architecture, the submodels of different sizes can reduce the computational overhead by more than 31% and 61%, respectively, while maintaining accuracy within an acceptable range.

In future work, we will integrate lightweight techniques to further reduce the computational load of HAR models, enabling their deployment on resource-constrained small devices. Furthermore, we will also further investigate the energy consumption of the model, allowing it to reduce energy usage while maintaining high accuracy.

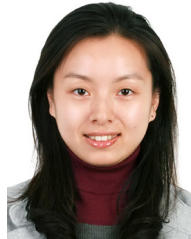
REFERENCES

- [1] E. Ramanujam, T. Perumal, and S. Padmavathi, "Human activity recognition with smartphone and wearable sensors using deep learning techniques: A review," *IEEE Sensors J.*, vol. 21, no. 12, pp. 13029–13040, Jun. 2021.
- [2] L. M. Dang, K. Min, H. Wang, M. J. Piran, C. H. Lee, and H. Moon, "Sensor-based and vision-based human activity recognition: A comprehensive survey," *Pattern Recognit.*, vol. 108, Dec. 2020, Art. no. 107561.
- [3] X. Zhou, W. Liang, K. I.-K. Wang, H. Wang, L. T. Yang, and Q. Jin, "Deep-learning-enhanced human activity recognition for Internet of Healthcare Things," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6429–6438, Jul. 2020.
- [4] M. Çiçek, "Wearable technologies and its future applications," *Int. J. Electr. Electron. Data Commun.*, vol. 3, no. 4, pp. 45–50, 2015.
- [5] Y. Zhang, W. Liang, X. Yuan, S. Zhang, G. Yang, and Z. Zeng, "Deep learning based abnormal behavior detection for elderly healthcare using consumer network cameras," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2414–2422, Feb. 2024.
- [6] S. Sun, T. Peng, H. Huang, Y. Wang, X. Zhang, and Y. Zhou, "IoT motion tracking system for workout performance evaluation: A case study on dumbbell," *IEEE Trans. Consum. Electron.*, vol. 69, no. 4, pp. 798–808, Nov. 2023.
- [7] H. Zheng, Y. Chen, Y. Li, Z. Wang, J. Chen, and X. Zhu, "High-accuracy and fine-granularity human activity recognition method based on body RFID skeleton," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 1040–1051, Feb. 2024.
- [8] H.-G. Kim, G. Y. Kim, and J. Y. Kim, "Music recommendation system using human activity recognition from accelerometer data," *IEEE Trans. Consum. Electron.*, vol. 65, no. 3, pp. 349–358, Aug. 2019.
- [9] K. Chen, D. Zhang, L. Yao, B. Guo, Z. Yu, and Y. Liu, "Deep learning for sensor-based human activity recognition: Overview, challenges, and opportunities," *ACM Comput. Surv.*, vol. 54, no. 4, pp. 1–40, 2021.
- [10] S. Münzner, P. Schmidt, A. Reiss, M. Hanselmann, R. Stiefelhagen, and R. Dürichen, "CNN-based sensor fusion techniques for multimodal human activity recognition," in *Proc. ACM Int. Symp. Wearable Comput.*, 2017, pp. 158–165.
- [11] M. Ullah, H. Ullah, S. D. Khan, and F. A. Cheikh, "Stacked LSTM network for human activity recognition using smartphone data," in *Proc. 8th Eur. Workshop Vis. Inf. Process. (EUVIP)*, 2019, pp. 175–180.
- [12] I. Dirgová Luptáková, M. Kubovčík, and J. Pospíchal, "Wearable sensor-based human activity recognition with transformer model," *Sensors*, vol. 22, no. 5, p. 1911, 2022.
- [13] Y. Matsubara, M. Levorato, and F. Restuccia, "Split computing and early exiting for deep learning applications: Survey and research challenges," *ACM Comput. Surv.*, vol. 55, no. 5, pp. 1–30, 2022.
- [14] S. Laskaridis, A. Kouris, and N. D. Lane, "Adaptive inference through early-exit networks: Design, challenges and directions," in *Proc. 5th Int. Workshop Embed. Mobile Deep Learn.*, 2021, pp. 1–6.
- [15] S. Tang et al., "You need multiple exiting: Dynamic early exiting for accelerating unified vision language model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 10781–10791.
- [16] N. Rashid, B. U. Demirel, and M. A. Al Faruque, "AHAR: Adaptive CNN for energy-efficient human activity recognition in low-power edge devices," *IEEE Internet Things J.*, vol. 9, no. 15, pp. 13041–13051, Aug. 2022.
- [17] M. G. Morshed, T. Sultana, A. Alam, and Y.-K. Lee, "Human action recognition: A taxonomy-based survey, updates, and opportunities," *Sensors*, vol. 23, no. 4, p. 2182, 2023.
- [18] Y. Zhao, R. Yang, G. Chevalier, X. Xu, and Z. Zhang, "Deep residual bi-directional LSTM for human activity recognition using wearable sensors," *Math. Problems Eng.*, vol. 2018, no. 1, 2018, Art. no. 7316954.
- [19] W. Jiang and Z. Yin, "Human activity recognition using wearable sensors by deep convolutional neural networks," in *Proc. 23rd ACM Int. Conf. Multimedia*, 2015, pp. 1307–1310.
- [20] F. J. Ordóñez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition," *Sensors*, vol. 16, no. 1, p. 115, 2016.
- [21] Y. Shavit and I. Klein, "Boosting inertial-based human activity recognition with transformers," *IEEE Access*, vol. 9, pp. 53540–53547, 2021.
- [22] S. Mahmud et al., "Human activity recognition from wearable sensor data using self-attention," in *Proc. ECAI*, 2020, pp. 1332–1339.
- [23] Y. Han, G. Huang, S. Song, L. Yang, H. Wang, and Y. Wang, "Dynamic neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7436–7456, Nov. 2022.
- [24] G. Huang et al., "Glance and focus networks for dynamic visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4605–4621, Apr. 2023.
- [25] Y. Han et al., "Dynamic perceiver for efficient visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 5992–6002.
- [26] J. Xin, R. Tang, J. Lee, Y. Yu, and J. Lin, "DeeBERT: Dynamic early exiting for accelerating BERT inference," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguist.*, 2020, pp. 2246–2251.
- [27] F. Ilhan, G. Su, and L. Liu, "ScaleFL: Resource-adaptive federated learning with heterogeneous clients," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 24532–24541.

- [28] S. Teerapittayanon, B. McDanel, and H.-T. Kung, "BranchyNet: Fast inference via early exiting from deep neural networks," in *Proc. 23rd Int. Conf. Pattern Recognit. (ICPR)*, 2016, pp. 2464–2469.
- [29] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
- [30] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [31] A. Reiss and D. Stricker, "Introducing a new benchmarked dataset for activity monitoring," in *Proc. 16th Int. Symp. Wearable Comput.*, 2012, pp. 108–109.
- [32] D. Roggen et al., "Collecting complex activity datasets in highly rich networked sensor environments," in *Proc. 7th Int. Conf. Netw. Sens. Syst. (INSS)*, 2010, pp. 233–240.
- [33] T. Stiefmeier, D. Roggen, G. Ogris, P. Lukowicz, and G. Tröster, "Wearable activity tracking in car manufacturing," *IEEE Pervasive Comput.*, vol. 7, no. 2, pp. 42–50, Apr.–Jun. 2008.
- [34] H. Haresamudram, D. V. Anderson, and T. Plötz, "On the role of features in human activity recognition," in *Proc. ACM Int. Symp. Wearable Comput.*, 2019, pp. 78–88.
- [35] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [36] N. Y. Hammerla, S. Halloran, and T. Plötz, "Deep, convolutional, and recurrent models for human activity recognition using wearables," 2016, *arXiv:1604.08880*.
- [37] V. S. Murahari and T. Plötz, "On attention models for human activity recognition," in *Proc. ACM Int. Symp. Wearable Comput.*, 2018, pp. 100–103.
- [38] M. Zeng et al., "Understanding and improving recurrent networks for human activity recognition by continuous attention," in *Proc. ACM Int. Symp. Wearable Comput.*, 2018, pp. 56–63.
- [39] H. Ma, W. Li, X. Zhang, S. Gao, and S. Lu, "AttnSense: Multi-level attention mechanism for multimodal human activity recognition," in *Proc. IJCAI*, 2019, pp. 3109–3115.
- [40] A. Jee, K. Agrawal, and S. Prakriya, "A coordinated direct AF/DF relay-aided NOMA framework for low outage," *IEEE Trans. Commun.*, vol. 70, no. 3, pp. 1559–1579, Mar. 2022.
- [41] A. Jee and S. Prakriya, "Performance of energy and spectrally efficient AF relay-aided incremental CDRT NOMA-based IoT network with imperfect SIC for smart cities," *IEEE Internet Things J.*, vol. 10, no. 21, pp. 18766–18781, Nov. 2023.



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